Revision history

Revision No.	Revised Item	Issued Date
1.0	Initial	Mar. 09.'06

MEMORY

32M(2M x 16bit) Asynchronous Static Random Access Memory
CompactCell® SRAM with ***Deep Power Down*** and ***Page Mode***

DESCRIPTION

The SV6P3217RFB is a high speed, low power Static Random Access Memory (SRAM) organized as 2,097,152 words by 16 bits.
The SV6P3217RFB is a Pseudo SRAM based on successfully proven *Silicon7 CompactCell® SRAM* which was specifically developed for cost sensitive, low power computing and communication applications such as mobile cellular phone handsets, personal digital assistants and other battery-operated consumer products.
SV6P3217RFB support special mode such as Deep Power Down and Page Mode for more effective performance in 3G mobile applications.

FEATURES

- Standard Asynchronous SRAM Interface
- Proven *Silicon7 CompactCell® SRAM* for high density, low power and cost sensitive applications
- Organization : 2M x 16Bit
- Power Supply Voltage : 1.7 ~ 1.95 V
- Deep Power Down : 5uA
- Page Size : 4,8,16 word
- Tri-state Output and TTL Compatible
- Includes an on-Chip temperature sensor for TCR
- SV6P3217RFB-1
  - Page access time : 25ns (4,8,16-word)
  - Partial Array Refresh (PAR)
  - Reduced Memory Size Mode (RMS)
  - Deep sleep mode
- SV6P3217RFB-2
  - Page mode
  - Direct DPD when /ZZ goes LOW.

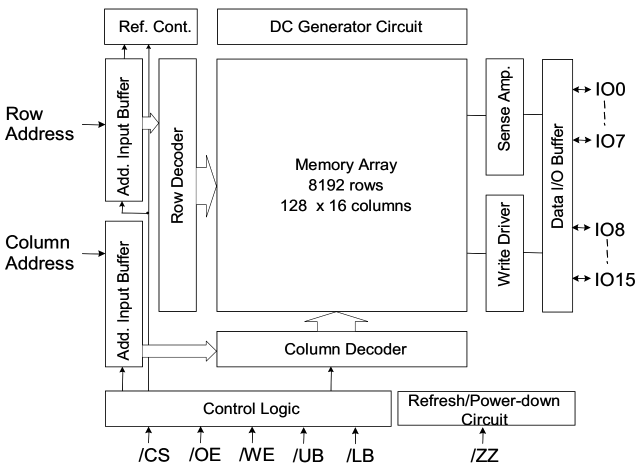
PRODUCT FAMILY

Product Family	Operating Temperature	Voltage	Speed	ISB1 (Max)	ICC1 (Max)	ICC2 (f=1MHZ,Max)	Mode
SV6P3217RFB-1	-25 ~ 85 ℃	1.7~1.95 V	70 ns	80uA	1.5mA	25mA	Page
SV6P3217RFB-2							

PIN DESCRIPTION

	1	2	3	4	5	6
A	/LB	/OE	A0	A1	A2	/ZZ
B	IO8	/UB	A3	A4	/CS	IO0
C	IO9	IO10	A5	A6	IO1	IO2
D	VSS	IO11	A17	A7	IO3	VCC
E	VCC	IO12	VSS	A16	IO4	VSS
F	IO14	IO13	A14	A15	IO5	IO6
G	IO15	A19	A12	A13	/WE	IO7
H	A18	A8	A9	A10	A11	A20

FUNCTIONAL BLOCK DIAGRAM



PRODUCT LIST

Part Name	Function
SV6P3217RFB-1 SV6P3217RFB-2	32M, FBGA , 70ns, 1.8V, -25℃～85℃ , Pb-free

ABSOLUTE MAXIMUM RATING

Item	Symbol	Ratings	Unit
Voltage on any pin relative to Vss	V_{IN}, V_{OUT}	−0.2 to VCC+0.3 V	V
Voltage on Vcc supply relative to Vss	V_{CC}	−0.2 to VCC+0.3V	V
Power Dissipation	P_D	1.0	W
Storage temperature	T_{STG}	−65 to 150	℃
Operating Temperature	T_A	−25 to 85	℃

Note : Stresses greater than those listed under “ABSOLUTE MAXIMUM RATINGS” may cause permanent damage to the device. This is stress rating only and the functional operation of the device under these or any other conditions above those indicated in the operation of this specification is not implied. Exposure to the absolute maximum rating conditions for extended period may affect reliability.

TRUTH TABLE [1,2]

MODE	/ZZ	/CS	/OE	/WE	/LB	/UB	IO[7:0]	IO[15:8]	Power
Direct DPD [3]	L	X	X	X	X	X	High-Z	High-Z	Deep Power Down
Deep Power Down [4]	L	H	X	X	X	X	High-Z	High-Z	
Deep Power Down [4]	L	X	X	X	H	H	High-Z	High-Z	
Deep Power Down [4]	L	H	X	X	H	H	High-Z	High-Z	
Deselected	H	H	X	X	X	X	High-Z	High-Z	Standby
Deselected	H	X	X	X	H	H	High-Z	High-Z	
Deselected	H	L	X	X	H	H	High-Z	High-Z	
Lower Byte Read	H	L	L	H	L	H	Data Out	High-Z	Active
Upper Byte Read	H	L	L	H	H	L	High-Z	Data Out	
Word Read	H	L	L	H	L	L	Data Out	Data Out	
Lower Byte Write	H	L	X	L	L	H	Data In	High-Z	
Upper Byte Write	H	L	X	L	L	H	High-Z	Data In	
Word Write	H	L	X	L	L	L	High-Z	Data In	

Note:
1. When /LB and /UB are in select mode (LOW), IO[15:0] are affected. When /LB only is in select mode, only IO[7:0] are affected. When /UB only is in the select mode, IO[15:8] are affected. If both /UB and /LB are in the deselect mode (HIGH), the chip is in a standby mode regardless of the state of /CS.
2. H = Logic HIGH, L= Logic Low, X = Don’t Care.
3. SV6P3217RFB-2: DPD mode when /ZZ goes LOW.
4. SV6P3217RFB-1: During /ZZ = L and /CS = H, Mode depends on how the VAR is set up either in PAR or Deep Power-Down Modes.

RECOMMENDED DC OPERATING CONDITIONS

Parameter	Symbol	Min	Typ	Max	Unit
Supply Voltage	Vcc	1.7	1.8	1.95	V
Ground	Vss	0	0	0	V
Input High Voltage	VIH	0.8 * Vcc	-	Vcc + 0.2	V
Input Low Voltage	VIL	-0.2	-	0.4	V

CAPACITANCE [5]

Symbol	Parameter	Condition	Max.	Unit
CIN	Input Capacitance	TA = 25℃, f = 1MHz, Vcc = Vcc(typ)	8	pF
COUT	Output Capacitance		8	pF

DC AND OPERATING CHARACTERISTICS [6]

Parameter	Symbol	Test Condition	Min	Max	Unit
Input Leakage Current	ILI	VIN = Vss to Vcc		1	uA
Output Leakage Current	ILO	/CS1 = VIH, CS2=VIL,/UB=/LB= VIH or /OE=VIH or /WE=VIL, VIO=Vss to Vcc		1	uA
Average Operating Current	Icc1	Cycle Time = 1 us, 100%duty, IIO=0mA, /CS1 ≤ 0.2V, CS2 ≥ Vcc-0.2V, VIN ≤ 0.2V or VIN ≥ Vcc-0.2V		5	mA
	Icc2	Cycle time=Min, IIO=0mA, 100% duty /CS1 = VIL, CS2=VIH,VIN=VIL or VIH		25	mA
Input Low Voltage	VIL		-0.2	0.2 * Vcc	V
Input High Voltage	VIH		0.8 * Vcc	Vcc + 0.3	V
Output Low Voltage	VOL	IOL = 0.2 mA		0.2Vcc	V
Output High Voltage	VOH	IOH = -0.2 mA	0.8Vcc		V
Standby Current(TTL)	ISB	/CS=VIH, /ZZ=VIL, /UB=/LB= VIH , Other inputs = VIH or VIL		0.3	mA
Standby Current(CMOS)	ISB2	/CS ≥ Vcc-0.2V, Other inputs = VIH or VIL		80	uA
Deep Power Down Current	Izz	/ZZ ≤ Vss+0.2V, /CS=VIH or /UB=/LB= VIH		5	uA

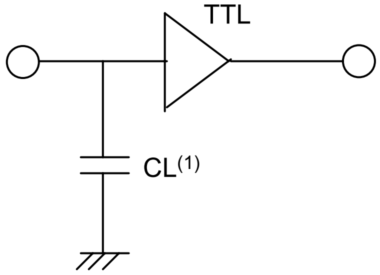
Note : 5.Overshoot and undershoot are sampled, not 100% tested.
6.Tested initially and after any design or process changed that may affect these parameters

AC TEST CONDITIONS [7]

T_A = -25℃ to 85℃ (Normal), unless otherwise specified

PARAMETER	Value
Input Pulse Level	0.1Vcc to 0.9Vcc
Input Rise and Fall Time	1V / ns
Input and Output Timing Reference Level	0.5Vcc
Output Load	CL = 30pF + 1TTL Load

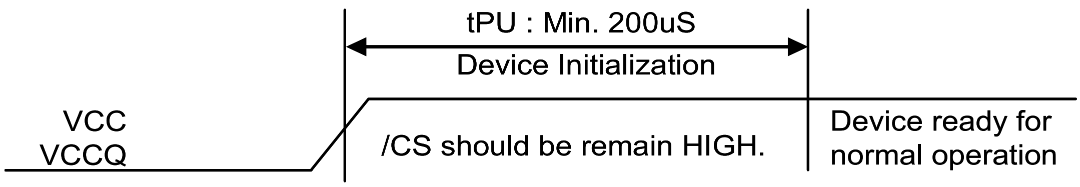
AC TEST LOADS



Note : 7. Including jig and scope capacitance

POWER UP Sequence

The initialization sequence is shown in the figure below. Chip Select (/CS) should be HIGH for at least 200 μs after VCC has reached a stable value. No access must be attempted during this period of 200 μs.



Low-Power Modes [8]

At power-up, all four sections of the die are activated and the PSRAM enters into its default state of full memory size and refresh space. This device provides four different Low-Power Modes

1. Reduced Memory Size Operation
2. Partial Array Refresh
3. Deep Power-Down Mode
4. Temperature Compensated Refresh

Reduced Memory Size Operation

In this mode of operation, the 16Mb PSRAM can be operated as a 24Mb, 16Mb or a 8Mb device. Please refer to “Variable Address Space Register (VAR)” on page 5 for the protocol to turn on/off sections of the memory. The device remains in RMS mode until changes to the VAR are made to revert back to a complete 16Mb PSRAM.

Partial Array Refresh

The Partial Array Self Refresh mode allows customers to turn off sections of the memory block in the Stand-by mode (with /ZZ tied LOW) to reduce standby current. In this mode the PSRAM will only refresh certain portions of the memory in the Stand-By Mode, as configured by the user through the settings in the Variable Address Register.
Once /ZZ returns HIGH in this mode, the PSRAM goes back to operating in full address refresh. Please refer to “Variable Address Space Register” on page 5 for the protocol to turn off sections of the memory in Stand-By mode. If the VAR register is not updated after the power up, the PSRAM will be in its default state. In the default state the whole memory array will be refreshed in the Stand-By Mode.

Deep Power-Down Mode

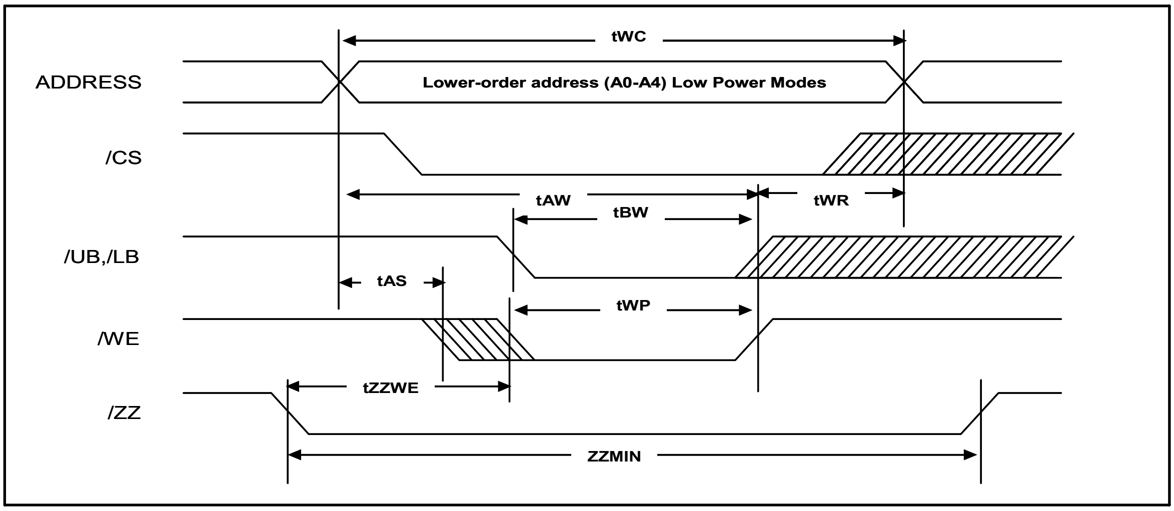
In this mode, the data integrity in the PSRAM is not guaranteed. This mode can be used to lower the power consumption of the PSRAM in an application. This mode can be enabled and disabled through VAR similar to the RMS and PAR mode. Deep Power-Down Mode is activated by driving /ZZ LOW. The device stays in the deep sleep mode until /ZZ is driven HIGH..

Temperature Compensated Refresh [9]

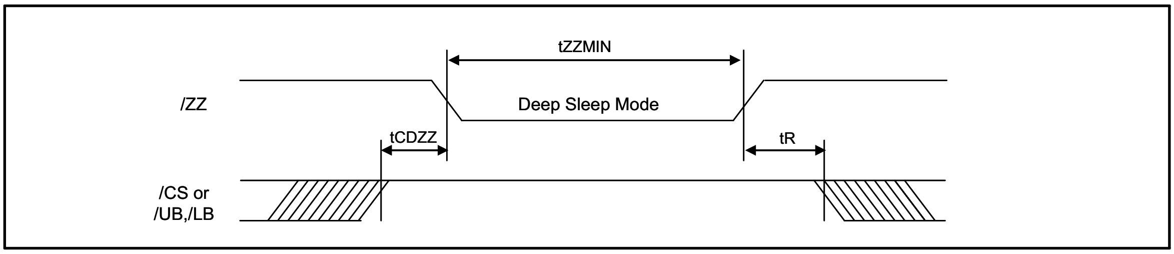
Temperature compensated refresh (TCR) allows for adequate refresh at different temperatures. This SV6P3217RFB-X device includes an on-chip temperature sensor. When the sensor is enabled, it continually adjusts the refresh rate according to the operating temperature. The on-chip sensor is enabled by default.

Note : (8) SV6P3217RFB-1 supports various low power modes by VAR update
(9) Both SV6P3217RFB-1 and SV6P3217RFB-2 support TCR

Variable Address Mode Register (VAR) update [8,10]



Deep Power-Down Mode – Entry/Exit Timings [8,11]



Variable Address and Deep Power-Down Timings

Description	Symbol	Min	Max	Unit
/ZZ LOW to /WE LOW	tZZWE		1	us
Write register cycle time	tWC	70		ns
Chip enable to end of write	tCW	70		ns
/UB,/LB low to end of write	tBW	70		ns
Address valid to end of write	tAW	70		ns
Write Recovery time	tWR	0		ns
Address setup time	tAS	0		ns
Write pulse width	tWP	40		ns
Chip deselect to /ZZ LOW	tCDZZ	5		ns
Deep Power-Down Pulse width	tZZMIN	8		us
Deep Power-Down Recovery	tR	200		us

Note : 10) /OE and the data pins are in a don't care state while the device is in variable address mode
11) tR applies only in the deep power-down mode.

Variable Address Space Register (VAR) [8]

A20 – A5	A4	A3	A2	A1	A0

A20 – A5: Reserved

CR[4]	ZZb Enable/Disable
0	Deep Sleep Enable (Default)
1	Deep Sleep Disable

CR[3]	Array On/Off on ZZb
0	PAR Mode (Default)
1	RMS Mode

CR[2]	Top/Bottom Half Selection
0	Bottom (Default)
1	Top

CR[1]	CR[0]	Memory Array Selection
0	0	Full Array (Default)
1	0	1/2 array (16M)
1	1	1/4 array (8M)
0	1	3/4 array (24M)

Variable Address Space – Address Patterns

Partial Array Refresh Mode (A[3] = 0, A[4] = 1)

A2	A1, A0	Active Section	Address Space	Size	Density
0	1 1	Top 1/4th of die	000000h-07FFFFh (A20 = A19 = 0)	512K x 16	8M
0	1 0	Top 1/2th of die	000000h-0FFFFFh (A20 = 0)	1M x 16	16M
0	0 1	Top 3/4th of die	000000h-17FFFFh (A20:A19 not equal to 1 1)	1.5M x 16	24M
1	1 1	Bottom 1/4th of die	180000h-1FFFFFh (A20 = A19 = 1)	512K x 16	8M
1	1 0	Bottom 1/2th of die	100000h-1FFFFFh (A20 = 1)	1M x 16	16M
1	0 1	Bottom 3/4th of die	080000h-1FFFFFh (A20:A19 not equal to 0 0)	1.5M x 16	24M

Reduced Memory Size Mode (A[3] = 1, A[4] = 1)

A2	A1, A0	Active Section	Address Space	Size	Density
0	1 1	Top 1/4th of die	000000h-07FFFFh (A20 = A19 = 0)	512K x 16	8M
0	1 0	Top 1/2th of die	000000h-0FFFFFh (A20 = 0)	1M x 16	16M
0	0 1	Top 3/4th of die	000000h-17FFFFh (A20:A19 not equal to 1 1)	1.5M x 16	24M
0	0 0	Full array	000000h-FFFFFFh (Default)	2M x 16	32M
1	1 1	Bottom 1/4th of die	180000h-1FFFFFh (A20 = A19 = 1)	512K x 16	8M
1	1 0	Bottom 1/2th of die	100000h-1FFFFFh (A20 = 1)	1M x 16	16M
1	0 1	Bottom 3/4th of die	080000h-1FFFFFh (A20:A19 not equal to 0 0)	1.5M x 16	24M
1	0 0	Full array	000000h-FFFFFFh (Default)	2M x 16	32M

Page Read/Write Mode

SV6P3217RFB-X device can be operated in a page read/write mode. This is accomplished by initiating a normal read/write of the device.
In order to operate the device in page mode, the upper order address bits should be fixed for four-word page access operation, all address bits except for A1 and A0 should be fixed until the page access is completed. For an eight-word page access, all address bits, except for A2, A1, and A0 should be fixed. For a sixteen-word page mode all address bits, except for A3, A2, A1, and A0, should be fixed.
The supported page lengths are four, eight, and sixteen words. Random page read/write is supported for all three four, eight, and sixteen-word page read/write options. Therefore, any address can be used as the starting address.

Page Mode Feature	4-Word Mode	8-Word Mode	16-Word Mode
Page Length	4 words	8 words	16 words
Page Read/Write Corresponding Addresses	A1, A0	A2, A1, A0	A3, A2, A1, A0
Page Read/Write Start Address	Don't Care	Don't Care	Don't Care
Page Direction	Don't Care	Don't Care	Don't Care

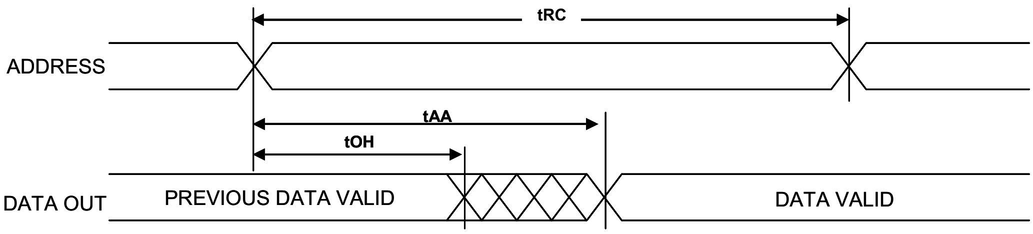
AC CHARACTERISTICS (Vcc = 1.7 ~ 1.95V, TA = -25 to 85℃) [12,13]

Description		Symbol	70ns		Unit
			Max	Max	
READ	Read Cycle Time	tRC	70		ns
	Address to data valid	tAA		70	ns
	/CS LOW to data valid	tCO		70	ns
	/OE LOW to data valid	tOE		35	ns
	/UB./LB Low to data valid	tBA		70	ns
	/CS LOW to Low-Z	tLZ	5		ns
	/UB, /LB to Low-Z	tBLZ	5		ns
	/OE LOW to Low-Z	tOLZ	5		ns
	/CS HIGH to High-Z	tHZ		25	ns
	/UB,/LB HIGH to High-Z	tBHZ		25	ns
	/OE HIGH to High-Z	tOHZ		25	ns
	Data hold from address change	tOH	5		ns
	Chip deselect time (pre-charge time) /CS, /UB&/LB HIGH pulse time	tCPH	5		ns
	Page Read Cycle Time	tPRC	25	20000	ns
	Page Read Address Access Time	tPAA		25	ns
WRITE	Write Cycle Time	tWC	70		ns
	/CS LOW to write end	tCW	60		ns
	Address Set-up to write end	tAW	60		ns
	/UB,/LB LOW to write end	tBW	60		ns
	/WE pulse width	tWP	50		ns
	Address Hold from write end	tWR	0		ns
	Address set-up to write start	tAS	0		ns
	/WE LOW to High-Z	tWHZ		25	ns
	Data set-up to write end	tDW	25		ns
	Data Hold from Write end	tDH	0		ns
	/WE HIGH to Low-Z	tOW	10		ns
	Chip deselect time (pre-charge time) /CS,/UB&/LB HIGH pulse time	tCPH	5		ns
	Page mode write cycle time	tPWC	25	20000	ns
	Address Hold from page write end	tPDW	25		ns
	Address set-up to page write start	tPDH	0		ns

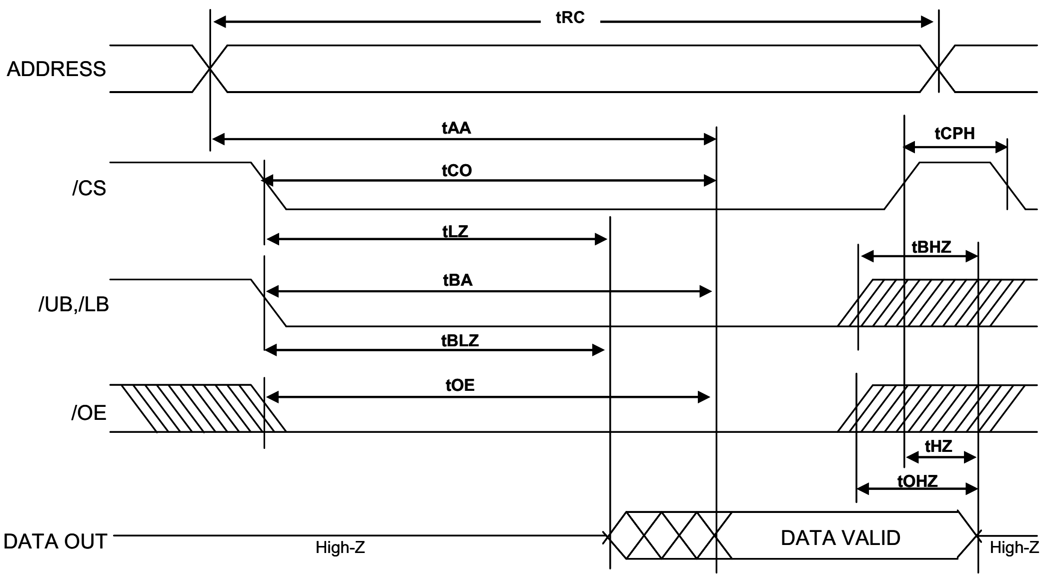
Note.
12. High-Z and Low-Z parameters are characterized and are not 100% tested.
13. In Page mode operation, the /CS pin must not be kept low longer than 20us. Otherwise ADD(4~20) must change at least within 20us.

TIMING DIAGRAMS

Read Cycle 1 (Address Transition Controlled) [14, 15]



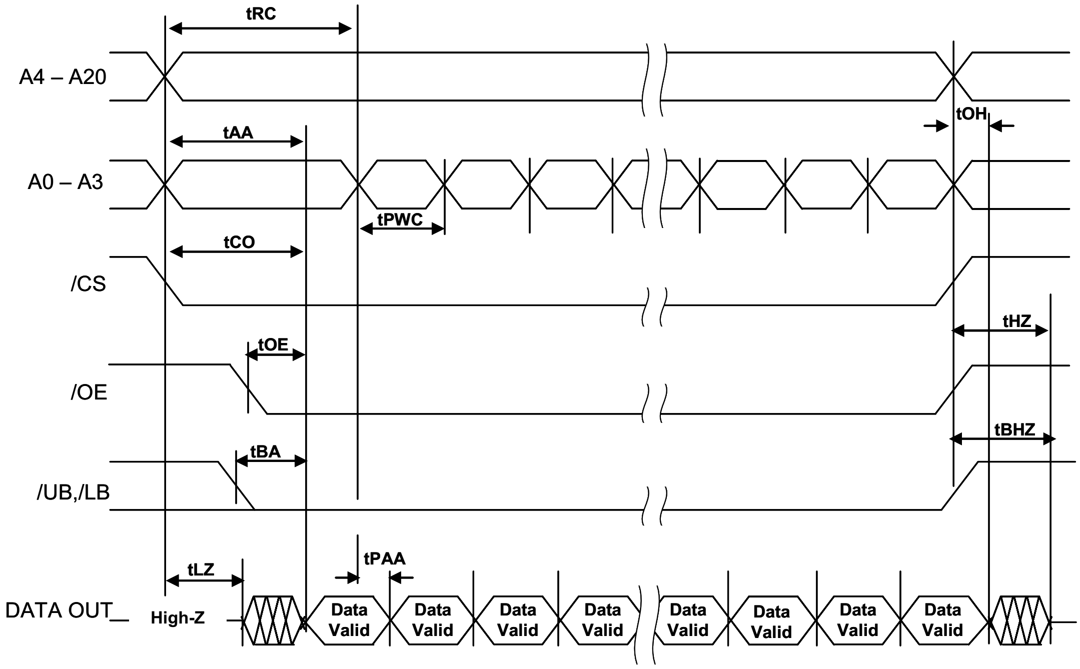
Read Cycle 2 (/OE Controlled) [15]



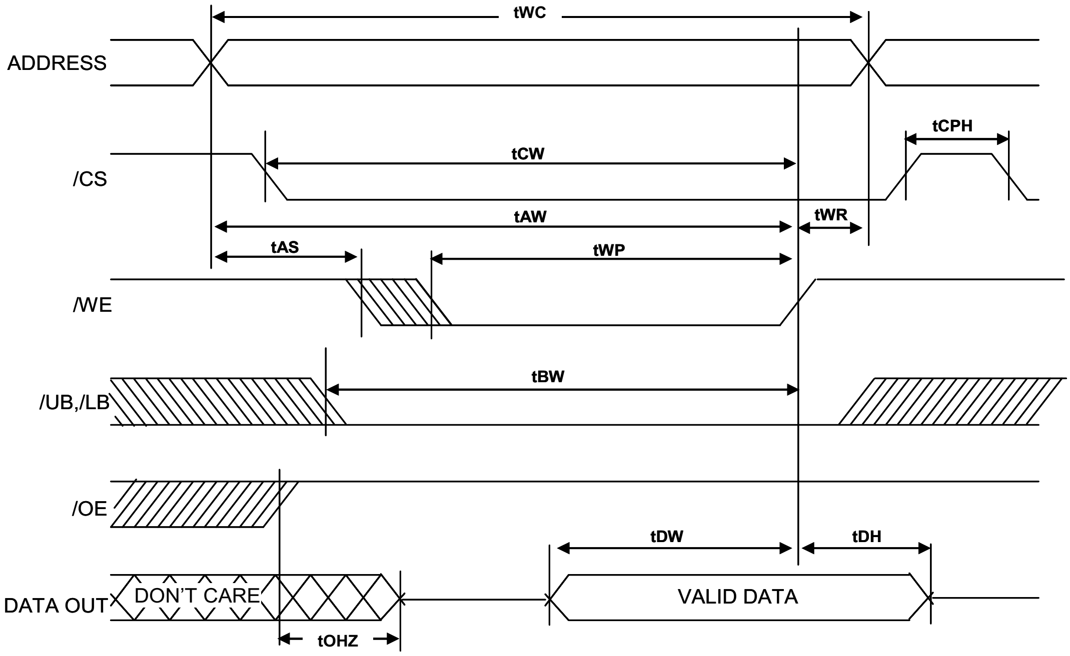
Note :
14. Device is continuously selected. /OE, /CE = VIL.
15. /WE is HIGH for Read cycle.

TIMING DIAGRAMS

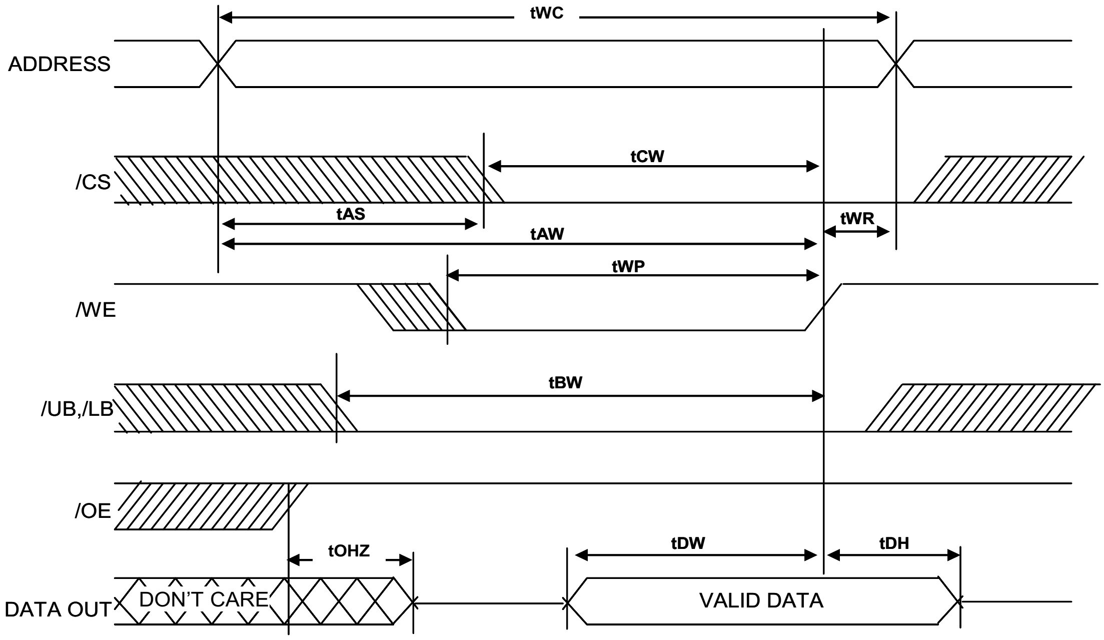
Page Read Cycle (/ZZ = /WE = VIH, 16 word access) [13, 15]



Write Cycle 1 (/WE controlled) [16, 17]

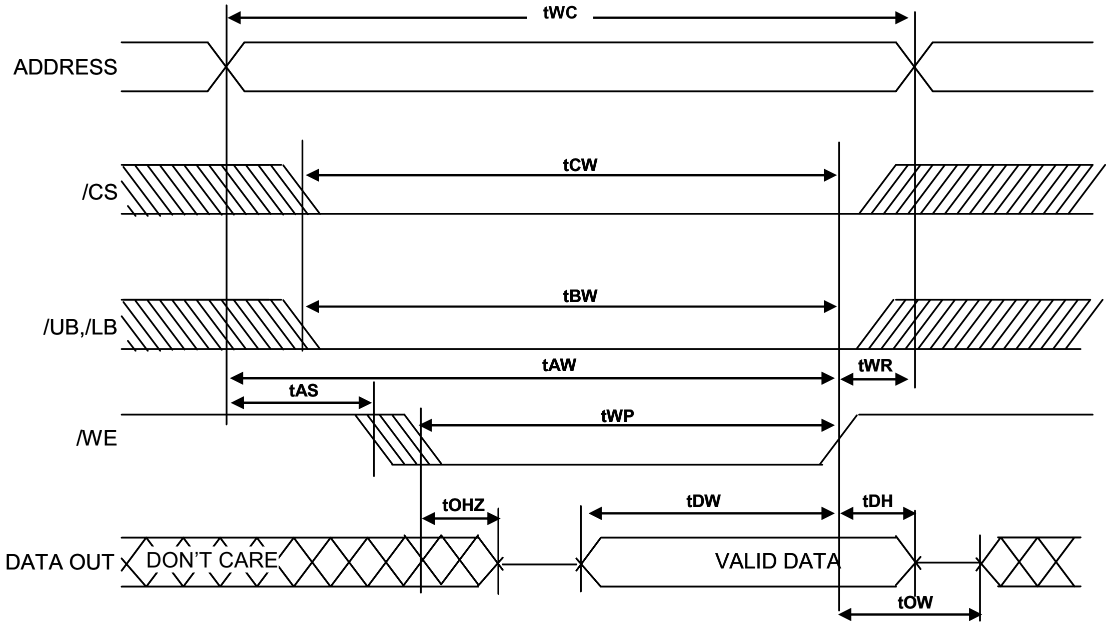


Write Cycle 2 (/CS controlled)

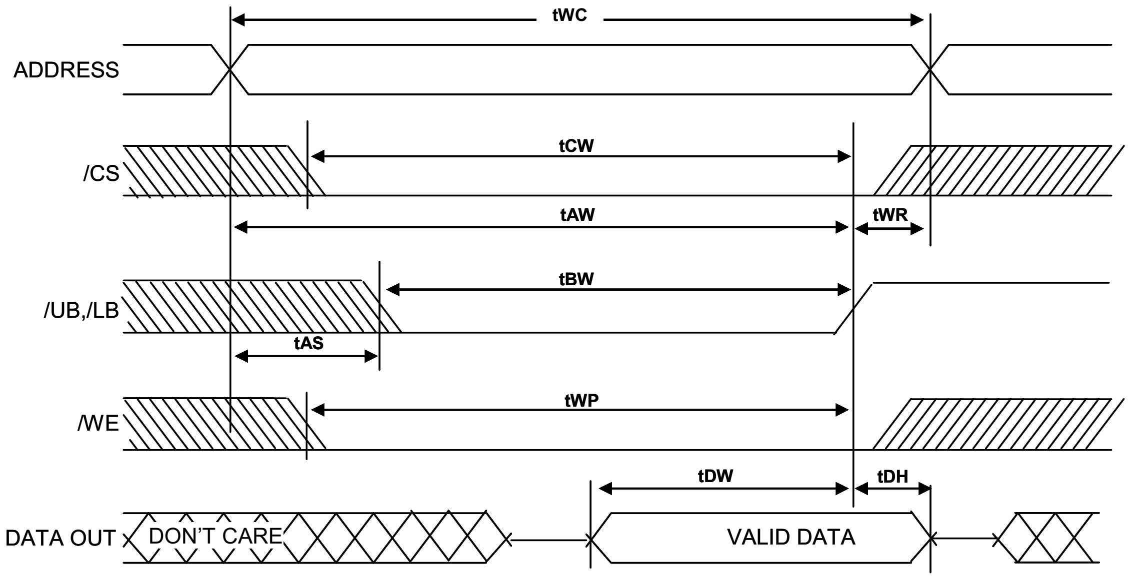


Notes (WRITE CYCLE) :
16. Data I/O is high-impedance if /OE ≥ VIH
17. During the DON'T CARE period in the DATA I/O waveform, the I/Os are in output state and input signals should not be applied.

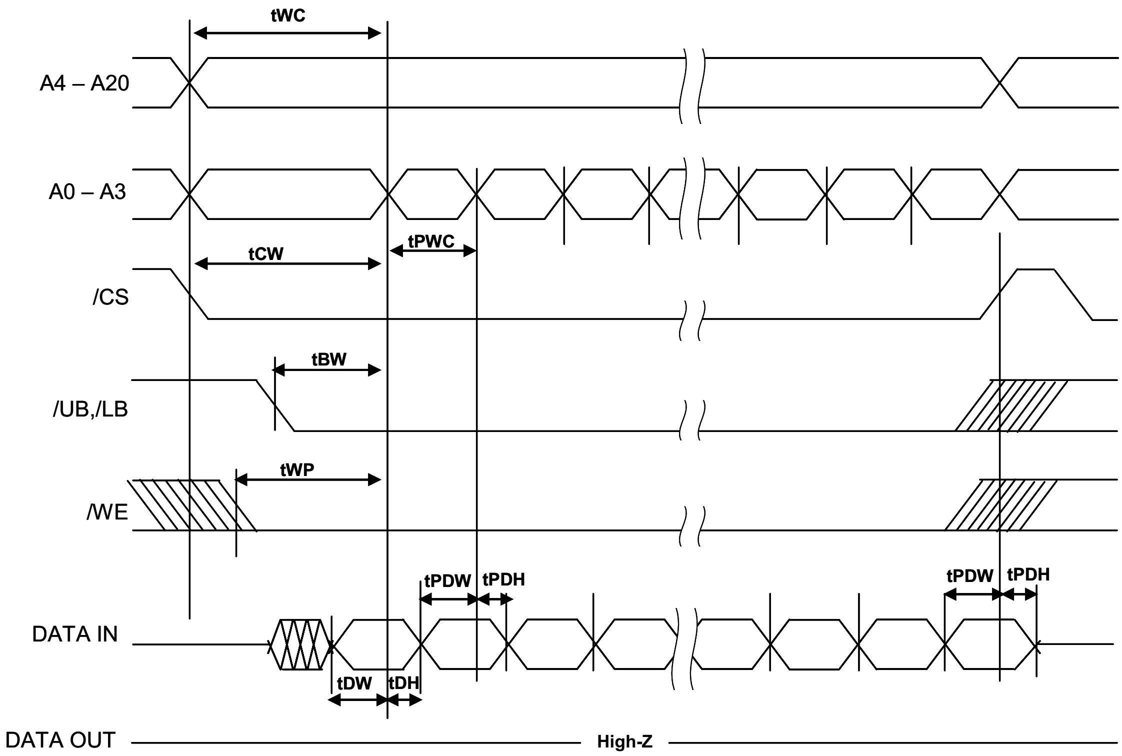
Write Cycle 3 (/WE controlled, /OE LOW)



Write Cycle 4 (/UB, /LB controlled, /OE LOW)



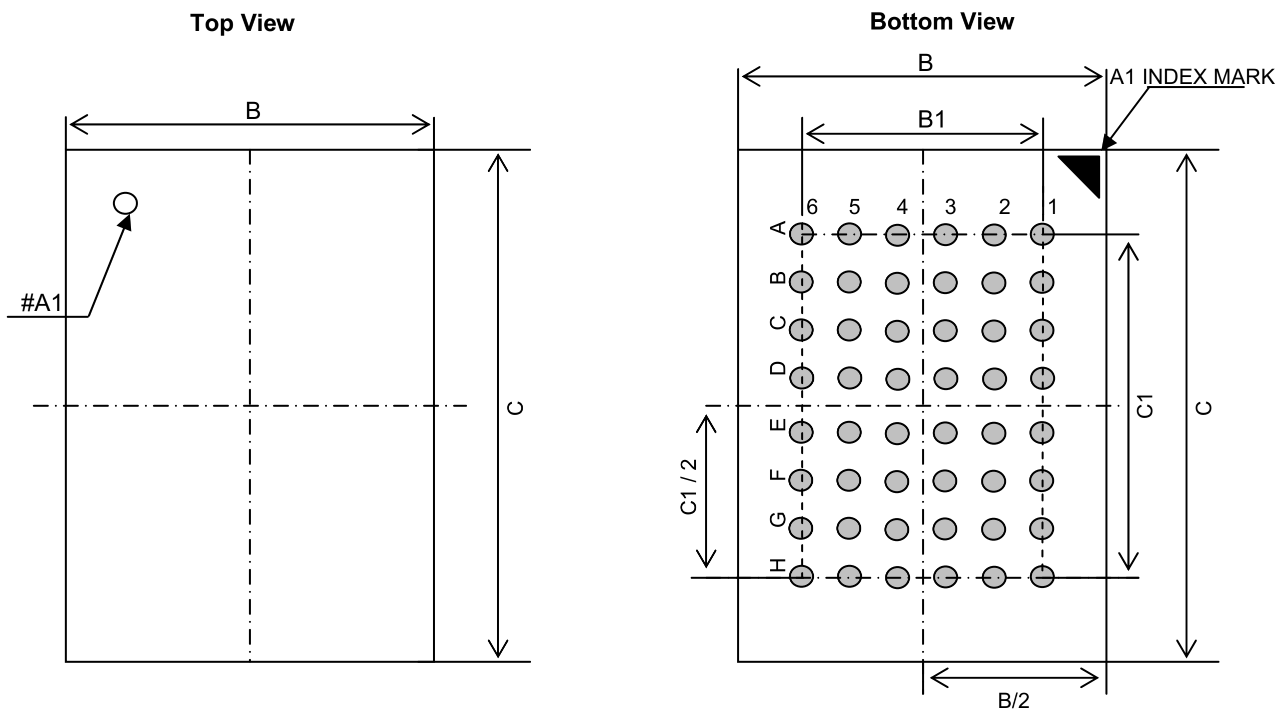
PAGE WRITE CYCLE (/ZZ = V_{IH} ,16 word access) [13]

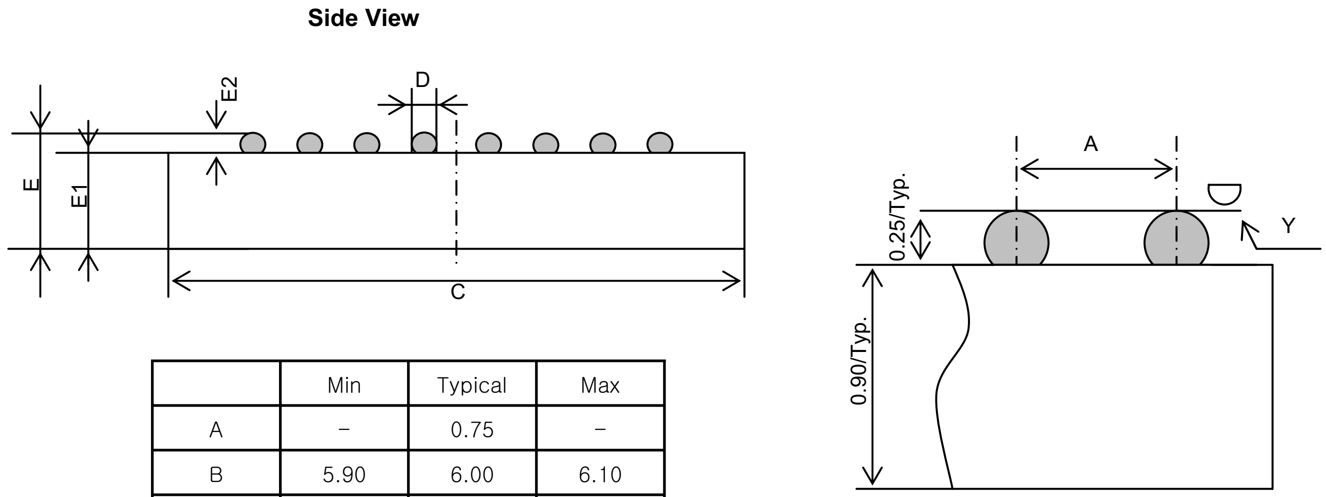


PACKAGE DIMENSION FOR BGA TYPE

Unit: millimeters

48 BALL FINE PITCH 6mm x 8mm BGA(0.75mm ball pitch)





	Min	Typical	Max
A	–	0.75	–
B	5.90	6.00	6.10
B1	–	3.75	–
C	7.90	8.00	8.10
C1	–	5.25	–
D	0.30	0.35	0.40
E	–	–	1.20
E1	–	–	0.90
E2	0.20	0.25	0.30
Y	–	–	0.10